PROJECT 17 · PHASE 1 · MACHINE LEARNING

# Where will the fire be tomorrow?

Next-day wildfire spread prediction with engineered tabular features: a leakage-aware evaluation on NDWS

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THE TASK

# For every 1 km pixel near an active fire, predict whether it burns tomorrow

**18,545**

patches of 64 × 64 km, US 2012–2020 (Kaggle NDWS, Huot et al. 2022)

**12**

input layers: terrain, weather, drought, vegetation, population, fire at day t

**1.1%**

of pixels are positive, so accuracy is useless: we report PR-AUC

Phase 1 = tabular ML: XGBoost / LightGBM on engineered features. Phase 2 = spatial deep learning on the same splits.

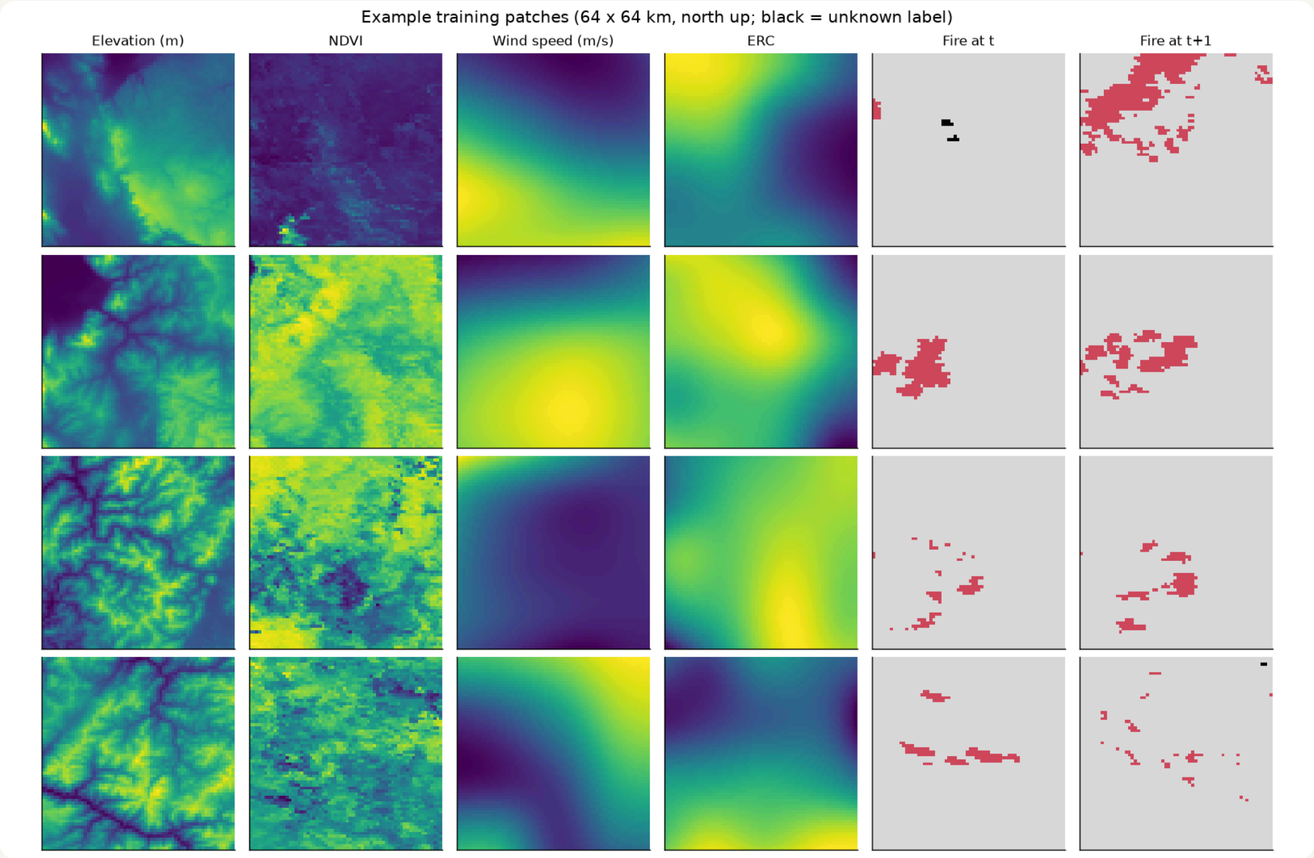
THE TEAM

# Each member owns one module end to end

| Member | Module | Code |
|---|---|---|
| Vedant Satbhai | Data access, spatial leakage audit | ndws.py, splits.py |
| Pratik Agade | EDA and data quality | eda.py, notebook 01 |
| Lakshay Saharan | Feature engineering | features.py, tests |
| Raghvendra Singh | Models, imbalance, tuning | dataset.py, train.py |
| Aaryan Krishna | Evaluation, interpretation, report | metrics.py, evaluate.py |

DATA

# Each sample: inputs at day t, fire mask at day t+1




**Official split:** 14,979 / 1,877 / 1,689 patches, by random weeks

**Labels:** 1 fire · 0 no fire · −1 unknown (cloud), excluded

**Cached** as NumPy (4.3 GB) for fast, repeatable runs

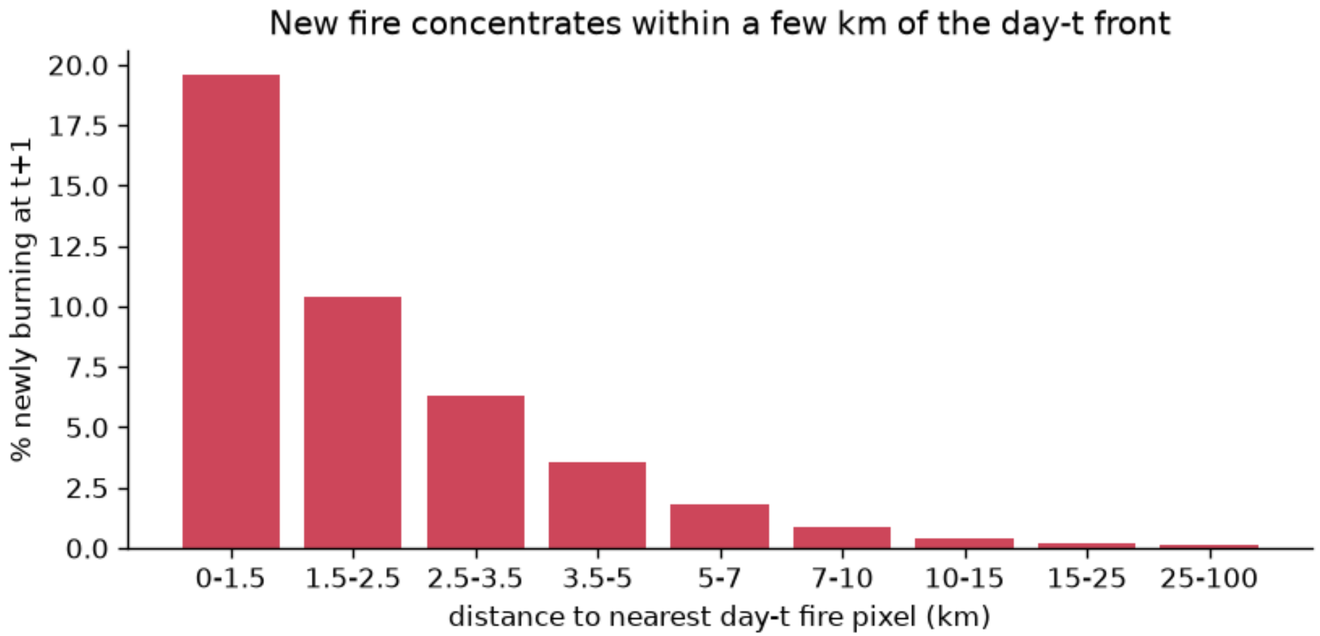
DATA QUALITY

# The raw data contains physically impossible values

| Found in the audit | Handling |
|---|---|
| Temperatures of 0 K and up to 344 K (71 °C) | Outside 200–330 K → missing |
| Wind directions from −496,522° to 37,620° | Outside 0–361° → missing |
| Negative humidity and wind speed | → missing |
| Negative precipitation (interpolation) | Clipped at 0 |
| Elevation slightly below sea level | Kept: real terrain |
| NDVI $\times 10^4$, heavy-tailed population | Rescaled, log1p |

EDA

# Most of tomorrow’s fire is new, and it starts near today’s front




**74%**

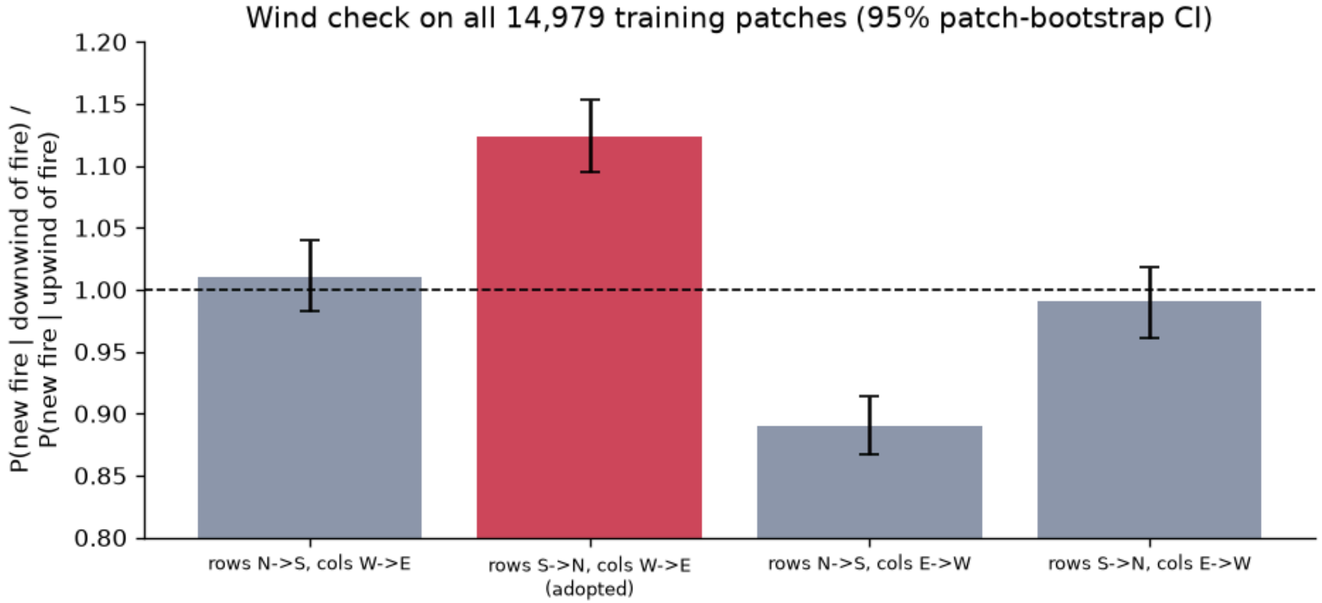
of day t+1 fire pixels were not burning at day t, so copying today’s mask fails

**20%**

ignition rate within 1.5 km of the front, vs 0.4% at 10–15 km

A PHYSICS TEST

# The data has no map orientation, so we let the wind decide



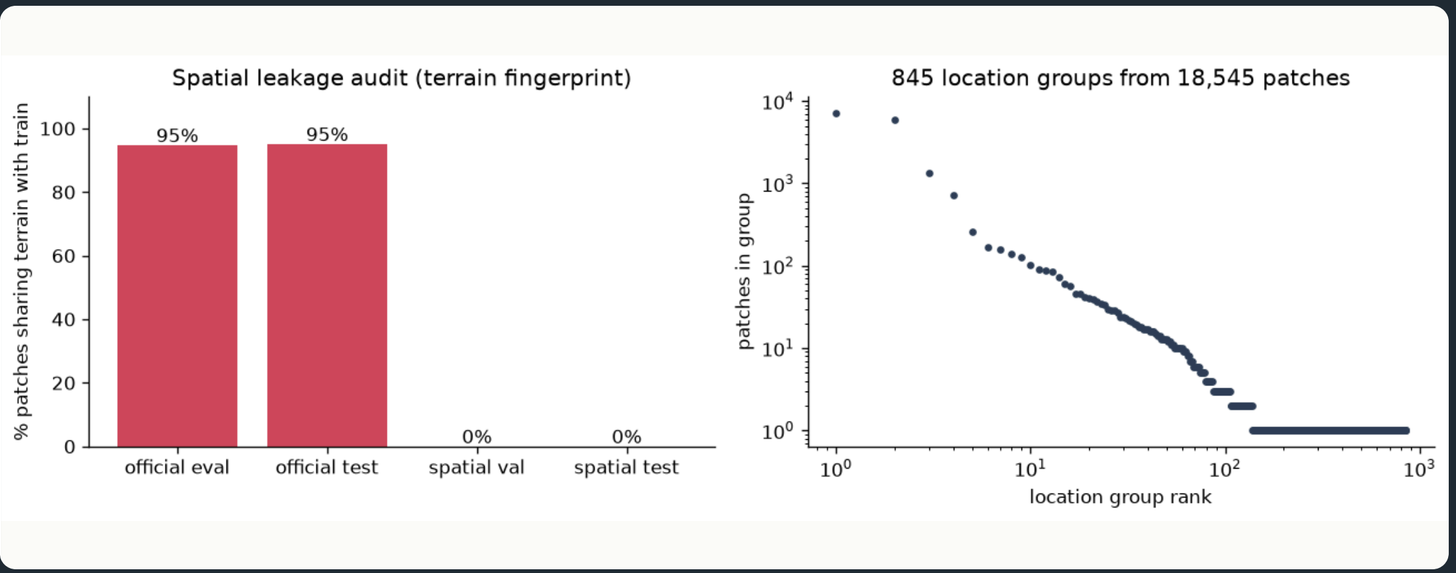

Fire should spread **downwind**. We tested all four row/column conventions on 14,979 patches.

Only rows south→north gives a real effect: **1.12×** (95% CI 1.09–1.15).

The usual assumption gives 1.01×: no wind effect at all.

SPATIAL LEAKAGE AUDIT

# 95% of official test patches cover the same ground as training




**Fingerprint:** elevation never changes, so an identical 8×8 km terrain block means the same place.

**Union-find** → 845 location groups. Control (mirrored): 0 / 1,855 matches.

**New split:** whole groups to train / val / test, 0% shared terrain.

FEATURE ENGINEERING

# 32 per-pixel features, all from day t only

### Fire (8)

burning share in 3/5/9/17 km windows, distance to fire, fire count

### Wind (6)

u, v; fire 1–3 km upwind or downwind; wind–fire alignment, wind push

### Terrain (4)

slope, aspect, rise over run from the nearest fire pixel

### Weather (2)

temperature range, vapour-pressure deficit (dryness)

### Vegetation (2)

5 × 5 km NDVI mean and spread (fuel amount, continuity)

### Raw (10)

cleaned elevation, weather, drought, NDVI, population, ERC

MODELS AND TRAINING

# Four learners, two baselines, tuned only on validation

### Models

Persistence and 3×3 persistence (no learning)
Logistic regression (L2)
Random forest
XGBoost, LightGBM with early stopping on val PR-AUC

### Imbalance

All positives + 2% of negatives, train split only (1.9 M rows). Probabilities corrected back:
**$p = r\cdot\hat{p} / (r\cdot\hat{p} + 1 - \hat{p}), r = 0.02$**

### A bug we fixed

First LightGBM stopped after 5 trees: pixels in a patch share weather, so small leaves memorise patches. Fix: leaves ≥ 2,000 rows, 50% column sampling.

RESULTS

# We beat the published tabular baselines and approach the CNN

| Test PR-AUC [95% CI] | Official split | Spatial split | Huot et al. |
|---|---|---|---|
| Persistence | 0.107 | 0.037 | 0.115 |
| Logistic regression | 0.230 | 0.117 | 0.198 |
| Random forest | 0.265 | 0.135 | 0.225 |
| **XGBoost** | **0.265** [0.248, 0.283] | **0.141** [0.128, 0.154] | – |
| LightGBM | 0.263 | 0.138 | – |
| CNN (Huot et al.) | – | – | 0.284 |

# Non-linearity matters.
# The choice of tree ensemble does not.

## +0.021

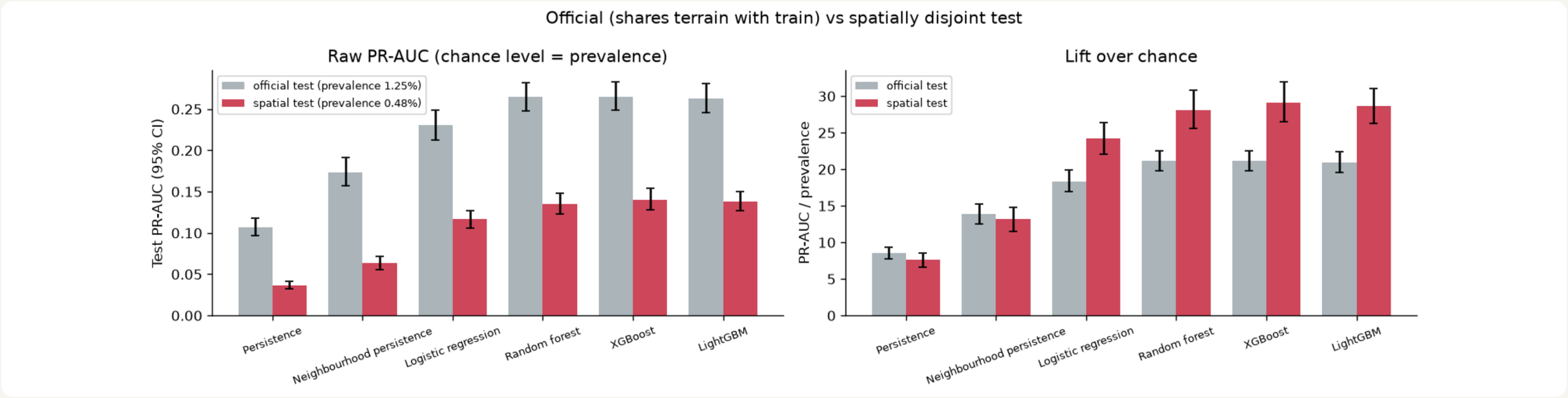
LightGBM over logistic regression (spatial), CI [+0.016, +0.026]: significant

## +0.003

LightGBM over random forest, CI [−0.002, +0.006]: not significant

DOES THE OVERLAP INFLATE THE BENCHMARK?

# Raw PR-AUC halves, but only because fires are rarer

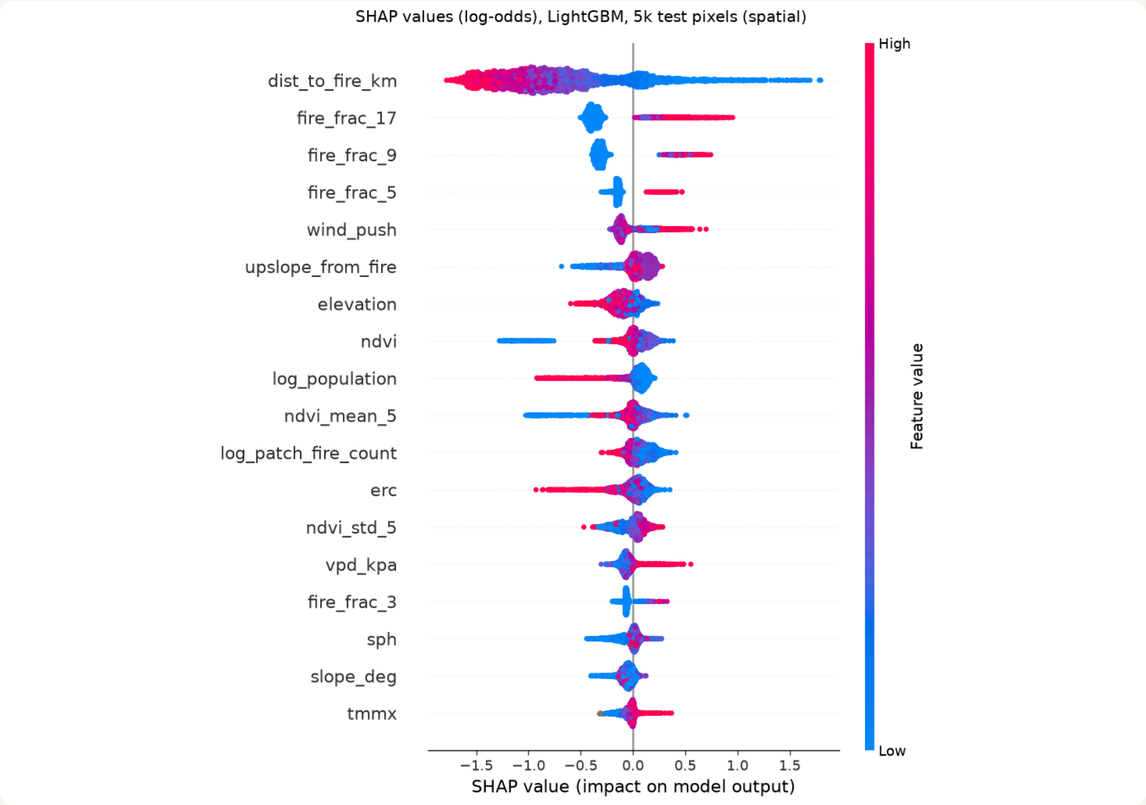



Spatial test prevalence 0.48% vs 1.25%. Lift over chance 28.6× vs 20.9×, ROC-AUC 0.896 vs 0.905, same model ranking: **no measurable inflation**.

WHAT DRIVES PREDICTIONS

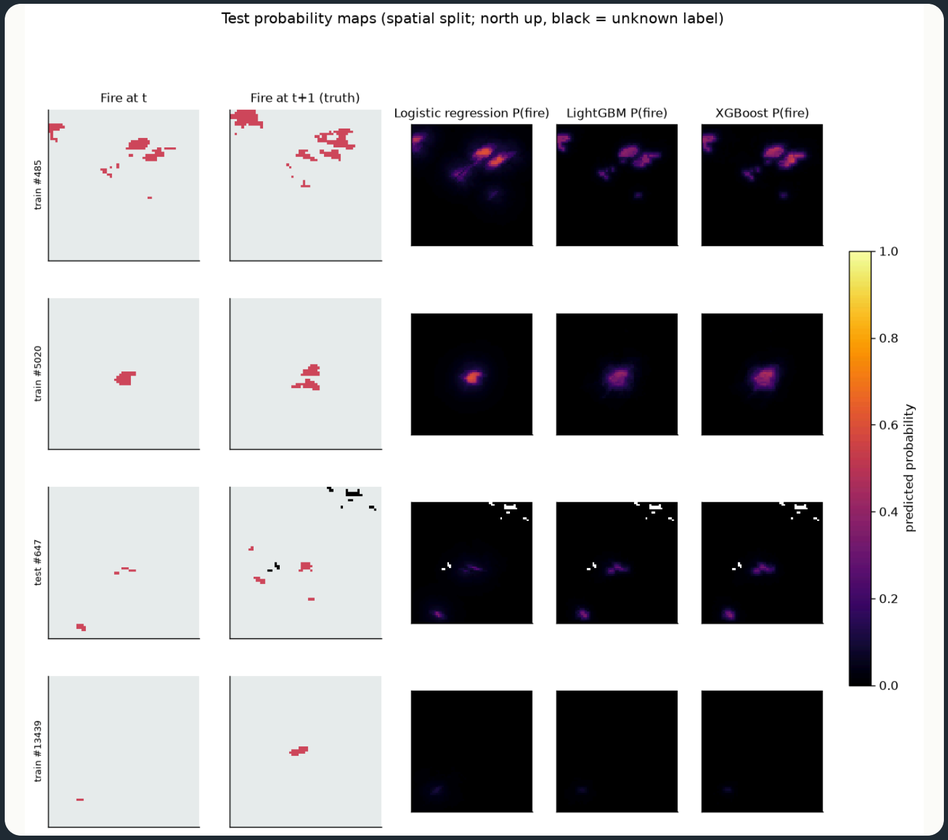
# Where the fire is now matters most; wind comes next

| LightGBM features | Test PR-AUC |
|---|---|
| Raw channels only | 0.010 |
| Fire features only | 0.111 |
| Raw + fire | 0.130 |
| + wind | 0.140 |
| All 32 | 0.138 |



PROBABILITY MAPS

# Tabular models draw halos, not fire fronts




Risk concentrates around the active front, and predicted rates match observed ones by distance (0.124 vs 0.117 within 1.5 km).

But blobs miss the front's shape and distant new ignitions (bottom row).

**That gap is what Phase 2's U-Net must close.**

LIMITATIONS AND NEXT STEPS

# What limits us, and what Phase 2 must prove

### Limitations

- Noisy daily MODIS labels: clouds, overpass timing
- No fuel moisture, fuel type, suppression or hourly wind
- Audit catches exact 8 × 8 km matches only
- Spatial test holds smaller fires than train

### Phase 2 plan

- U-Net on the 12-channel patch, same splits and metrics
- ConvLSTM only with multi-day data (WildfireSpreadTS)
- Hybrid: engineered channels or LightGBM output into the U-Net
- Must beat XGBoost on the spatial test, paired test

SUMMARY

Engineered fire and wind features + tree ensembles: **0.265 PR-AUC**, above published tabular baselines.

95% of the official test shares terrain with train, yet our results hold on a location-disjoint split.

# Questions?